

Modules and Vector Spaces

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1. Modules, vector spaces and algebras

We start off by recalling some basic language to describe sets endowed with structures.

Definition 1.1: For a non-empty set E , an *internal law of composition* is a map $\cdot : E \times E \rightarrow E$. Then the tuple $(E, \cdot)$ is referred to as a *magma*.

The properties of the law of composition give rise to various algebraic structures.

Definition 1.2:

- a. a magma $(E, \cdot)$ is a *semigroup* if $\cdot : E \times E \rightarrow E$ is associative;
- b. a semigroup $(E, \cdot)$ is a *monoid* if it has an identity element;
- c. a monoid $(E, \cdot)$ is a *group* if every element has an inverse.
- d. a group $(E, \cdot)$ is *abelian* if $\cdot : E \times E \rightarrow E$ is commutative.

A set may be endowed with multiple such composition laws, the interplay between which gives rise to further algebraic structures.

Definition 1.3: If E is a non-empty set, $+$: $E \times E \rightarrow E$ and $\cdot$: $E \times E \rightarrow E$ then the triplet $(E, +, \cdot)$ is a *ring* if

- a. $(E, +)$ is an abelian group;
- b. $(E, \cdot)$ is a monoid;
- c. $\forall x, y \in E$, we have $x(y + z) = xy + xz$ and $(y + z)x = xy + zx$.

If a ring has no nonzero zero divisors then it is an *integral domain*. If the non-zero elements of E form a group under $\cdot : E \times E \rightarrow E$ then it is a *field*.

Now, we encounter the central character of our narrative.

Definition 1.4: Let R be a ring. By a (*left*) R -*module*, we shall mean an abelian group M together with a left action $R \times M \rightarrow M$ described by $(\lambda, x) \mapsto \lambda x$ satisfying

- a. $\forall \lambda \in R$ and $\forall x, y \in M$ we have $\lambda(x + y) = \lambda x + \lambda y$;
- b. $\forall \lambda, \mu \in R$ and $\forall x \in M$ we have $(\lambda + \mu)x = \lambda x + \mu x$;
- c. $\forall \lambda, \mu \in R$ and $\forall x \in M$ we have $\lambda(\mu x) = (\lambda\mu)x$;
- d. $\forall x \in M$ we have $1_R x = x$.

The following properties are immediate.

Proposition 1.1: Let M be an R -module. Then

- a. $\forall \lambda \in R$ we have $\lambda 0_M = 0_M$;
- b. $\forall \lambda x \in M$ we have $0_R x = 0_M$;
- c. $\forall \lambda \in R$ and $\forall x \in M$ we have $\lambda(-x) = -(\lambda x) = (-\lambda)x$.

Moreover, if R is a division ring, then

- d. $\lambda x = 0_M$ implies that $\lambda = 0_R$ or $x = 0_M$.

Proof:

- a. $\lambda 0_M = \lambda(0_M + 0_M) = \lambda 0_M + \lambda 0_M$;
- b. $0_R x = (0_R + 0_R)x = 0_R x + 0_R x$;
- c. $\lambda(-x) + \lambda x = \lambda(-x + x) = \lambda 0_M = 0_M$ and $(-\lambda)x + \lambda x = (-\lambda + \lambda)x = 0_R x = 0_M$;
- d. Suppose $x \neq 0_M$. Then $\lambda^{-1} \lambda x = (\lambda)^{-1} 0_M = 0_M$.

□

Example:

- a. Every ring R is an R -module. Every field F is an F -vector space.
- b. Every additive abelian group M is a $\mathbb{Z}$ -module. The action $\mathbb{Z} \times M \rightarrow M$ is given by $m \mapsto mx$ where

$$mx = \begin{cases} \overbrace{x + x + \dots + x}^m & \text{if } m > 0; \\ 0 & \text{if } m = 0; \\ -|m|x & \text{if } m < 0. \end{cases}$$

- c. If R is a ring and $S \subseteq R$ is a subring containing 1_R then R is an S -module under the action $(s, r) \mapsto sr$.
- d. If R is a ring and $n \in \mathbb{N}_{>0}$ then the abelian group R^n of all n -tuples of elements of R under component-wise addition then R^n is a R -module under the left action $R \times R^n \rightarrow R^n$ defined by

$$r(x_1, \dots, x_n) = (rx_1, \dots, rx_n).$$

If F is a field then F^n is an F -vector space.

- e. Let R be a ring and $R^{\mathbb{N}}$ denote the set of all mappings $\mathbb{N} \rightarrow R$. We endow $R^{\mathbb{N}}$ with the addition

$$(f + g)(n) = f(n) + g(n).$$

$(R^{\mathbb{N}}, +)$ is an abelian group. We also define an action $R \times R^{\mathbb{N}} \rightarrow R^{\mathbb{N}}$ by $(r, f) \mapsto rf$ where rf is defined by

$$(rf)(n) = rf(n).$$

Then $(R^{\mathbb{N}}, +, \times)$ is an R -module.

Endowing a R -module with an appropriate multiplication operation gives rise to an R -algebra.

Definition 1.5: Let R be a commutative ring. An R -algebra is an R -module A together with an internal law of composition $A \times A \rightarrow A$, called *multiplication*, which is distributive over addition and such that

$$\forall \lambda \in R, \forall x, y \in A. \quad \lambda(xy) = (\lambda x)y = x(\lambda y).$$

If every non-zero element has an inverse, then it is a *division algebra*.

Example:

- a. If R is a ring then $\text{Mat}_{n \times n}(R)$, the set of $n \times n$ matrices over R , is an associative R -algebra.
- b. $\mathbb{C}$ is a division algebra over $\mathbb{R}$.
- c. Let R be a commutative ring and consider the R -module $R^{\mathbb{N}}$ defined prior. Given $f, g : \mathbb{N} \rightarrow R$, we define the product map $fg : \mathbb{N} \rightarrow R$ by

$$(fg)(n) = \sum_{i=1}^n f(i)g(n-i).$$

Then this multiplication makes $R^{\mathbb{N}}$ a R -algebra. This R -algebra is called the *algebra of formal power series with coefficients in R* . We justify this terminology now. First, define $t \in R^{\mathbb{N}}$ by

$$t(n) = \begin{cases} 1 & \text{if } n = 1; \\ 0 & \text{otherwise.} \end{cases}$$

and for every $m \in \mathbb{N}_{>0}$, the m -fold composite map

$$t^m = \underbrace{t \circ t \circ \dots \circ t}_m.$$

This admits a simple description

$$t^m = \begin{cases} 1 & \text{if } n = m; \\ 0 & \text{otherwise.} \end{cases}$$

We then consider the formal power series associated with $f \in R^{\mathbb{N}}$, given by

$$\nu = \sum_{i \geq 0} f(i)t^i,$$

where $t^0 = \text{id}_R$, the identity map on R . It is readily seen that $\forall n \in \mathbb{N}$, we have $f(n) = \nu(n)$.

2. Submodules, intersection and sums

We now discuss creating new modules out of old ones.

Definition 2.1: If S is a non-empty subset of an additive group G then S is said to be a *stable subset* of G , or *closed under the operation of G* , if

$$\forall x, y \in S. \quad x + y \in S.$$

It is clear that a stable subset is always a *semigroup*.

Definition 2.2: A *subgroup* of a group G is a non-empty subset that is stable and also a group under the induced law of composition.

We have a very clean characterization for subgroups.

Lemma 2.1: A non-empty subset $H \subseteq G$ of a group G is a subgroup if and only if $\forall x, y \in H$ we have $x - y \in H$.

Proof: The forward direction is clear. For the backward direction, we need to show the existence of the identity and inverses. For the first, note that for any $x \in H$, we get $0 = x - x \in H$. Next, for any $x \in H$ as $0 \in H$, we also have $-x = 0 - x \in H$. $\square$

These ideas also extend to modules and algebras.

Definition 2.3: By a *submodule* of an R -module M we mean a subgroup N of M that is stable under the action of R on M , in the sense that if $x \in N$ and $\lambda \in R$ then $\lambda x \in N$.

Lemma 2.2: A non-empty subset $N \subseteq M$ of an R -module M is a submodule of M if and only if $\forall x, y \in N$ and $\forall \lambda, \mu \in R$ we have $\lambda x + \mu y \in N$.

Definition 2.4: By a *subalgebra* of an R -algebra A we mean a submodule B of A that is closed under the multiplication law for A .

Lemma 2.3: A non-empty subset $B \subseteq A$ of an R -algebra A is a subalgebra if and only if

$$\forall x, y \in B. \quad \forall \lambda \in R. \quad x - y \in B, xy \in B, \lambda x \in B.$$

Example:

- a. Viewing a ring R as a R -module, its submodules are precisely the left ideals of R .

- b. Let $C([a, b])$ be the set of continuous function $[a, b] \rightarrow \mathbb{R}$. Then $C([a, b])$ is an $\mathbb{R}$ -vector space. The set $D([a, b])$ is the set of differentiable function $[a, b] \rightarrow \mathbb{R}$, which is a subspace of $C([a, b])$. Under point-wise multiplication, $C([a, b])$ is an $\mathbb{R}$ -algebra and $D([a, b])$ is a sub-algebra.
- c. If G is an abelian group then the submodules of the $\mathbb{Z}$ -module G are simply the subgroups of G .

The set of submodules, ordered by set inclusion, actually has a lattice structure.

Proposition 2.1: The intersection of any family of submodules of an R -module M is a submodule of M .

Definition 2.5: Let $S \subseteq M$ be a subset of the module M . The *submodules generated by S* , denoted $\langle S \rangle$, is the smallest submodule of M containing S .

It is possible to give an explicit description of $\langle S \rangle$.

Definition 2.6: Let M be an R -module and let S be a non-empty subset of M . Then $x \in M$ is a *linear combination of elements of S* if there exist elements $x_1, \dots, x_n$ in S and scalars $\lambda_1, \dots, \lambda_n$ in R such that

$$x = \sum_{i=1}^n \lambda_i x_i.$$

We denote the set of all linear combinations of elements of S by $\text{span}(S)$.

Proposition 2.2: Let S be a subset of the R -module M . Then

$$\langle S \rangle = \begin{cases} \{0\} & \text{if } S = \emptyset; \\ \text{span}(S) & \text{otherwise.} \end{cases}$$

Definition 2.7: We say that an R -module M is *generated* by the subset S , or that S is a *set of generators* of M , when $\langle S \rangle = M$. By a *finitely-generated R -module* we mean an R -module which has a finite set of generators.

Thus, set intersection is the infimum operation for our lattice. Now, we introduce the supremum operation.

Proposition 2.3: Let $(M_i)_{i \in I}$ be a family of submodules of an R -module M . If $\mathbb{P}^*(I)$ denotes the set of all non-empty finite subsets of I then the submodule generated by $\cup_{i \in I} M_i$ consists of all finite sums of the form $\sum_{j \in J} m_j$ where $J \in \mathbb{P}^*(I)$ and $m_j \in M_j$.

Definition 2.8: Let $(M_i)_{i \in I}$ be a family of submodules of an R -module M . We call the submodule generated by the family $(M_i)_{i \in I}$ the *sum of the family $(M_i)_{i \in I}$* and denote it $\sum_{i \in I} M_i$.

Lemma 2.4: The following properties of the sum of submodules is immediate.

- a. (commutativity of σ) If $\sigma : I \rightarrow I$ is a bijection then $\sum_{i \in I} M_i = \sum_{i \in I} M_{\sigma(i)}$.
- b. (associativity of σ) If $(I_k)_{k \in A}$ is a family of non-empty subsets of I with $I = \cup_{k \in A} I_k$ then

$$\sum_{i \in I} M_i = \sum_{k \in A} \left(\sum_{i \in I_k} M_i \right).$$

- c. $\forall i \in I$, we have

$$\sum_{i \in I} M_i = M_i + \sum_{j \neq i} M_j.$$

The lattice of submodules is actually modular.

Proposition 2.4: If M is an R -module and if A, B, C are submodules of M with $C \subseteq A$ then

$$A \cap (B + C) = (A \cap B) + C.$$

Proof: As $C \subseteq A$, we know that $A + C = A$. Note that $(A \cap B) + C \subseteq A + C = A$ and $(A \cap B) + C \subseteq B + C$. That is,

$$(A \cap B) + C \subseteq A \cap (B + C).$$

For the reverse inclusion, let $a \in A \cap (B + C)$ and $b \in B, c \in C$ such that $a = b + c$. Then, $b = a - c \in A + C = A$. Thus, $a = b + c \in (A \cap B) + C$. $\square$

3. Morphisms and exact sequences

We are interested in structure-preserving maps. For example, from the theory of groups,

Definition 3.1: Let G, H be groups. A map $f : G \rightarrow H$ is called a *morphism* (or *homomorphism*) if

$$\forall x, y \in G. \quad f(x + y) = f(x) + f(y).$$

Lemma 3.1: A morphism $f : G \rightarrow H$ maps G onto a subgroup of H ,

$$\text{Im } f = \{f(x) \mid x \in G\}.$$

Furthermore, we have

- a. $f(0_G) = 0_H$;
- b. $\forall x \in G. \quad f(-x) = -f(x)$.

We can extend the notion of a morphism to modules, in a manner that still preserves the two properties given by the lemma.

Definition 3.2: If M and N are R -modules then a mapping $f : M \rightarrow N$ is called an *R -morphism* if

- a. $\forall x, y \in M. \quad f(x + y) = f(x) + f(y)$;
- b. $\forall x \in M. \forall \lambda \in R. \quad f(\lambda x) = \lambda f(x)$. When R is a field, an R -morphism is called a *linear transformation*. An R -morphism is called an *R -monomorphism* if it is injective; an *R -epimorphism* if it is surjective; and an *R -isomorphism* if it is bijective. An R -morphism $f : M \rightarrow M$ is an *R -endomorphism* on M .

Example:

- a. For abelian groups M, N viewed as $\mathbb{Z}$ -modules, a $\mathbb{Z}$ -morphism is just a group morphism.
- b. If M is an R -module and $n \in \mathbb{Z}^+$, the mapping $\text{pr}_i : M^n \rightarrow M$ described by

$$\text{pr}_i(x_1, \dots, x_n) = x_i$$

is an R epimorphism, called the *i-th projection of M^n onto M* .

An important property of morphisms $f : M \rightarrow N$ is that it induces mappings between the corresponding lattices of submodules.

Proposition 3.1: Let M and N be R -modules and $f : M \rightarrow N$ an R -morphism. Then for every submodule X of M the set $f(X)$ is a submodule of N and for every submodule Y of N the set $f^{-1}(Y)$ is a submodule of M .

Particularly, this lets us define mappings $f : L(M) \rightarrow L(N)$ and $f^{-1} : L(N) \rightarrow L(M)$ wherein $L(\cdot)$ denotes the lattice of submodules. Note also that these induced mappings are inclusion-preserving. For an R -morphism $f : M \rightarrow N$, $f(M)$ is also called the *image* of f and $f^{-1}(\{0_N\})$ is called the *kernel* of f . Immediately, we have,

Lemma 3.2: Let $f : M \rightarrow N$ be an R -morphism. Then

- a. f is a monomorphism if and only if $\text{Ker}(f) = \{0_M\}$;
- b. f is an epimorphism if and only if $\text{Im}(f) = N$.

We now expand on some properties of the map induced on the lattice of submodules.

Lemma 3.3: Let $f : M \rightarrow N$ be an R -morphism. If A is a submodule of M and B is a submodule of N then

- a. $f(A \cap f^{-1}(B)) = f(A) \cap B$;
- b. $f^{-1}(B + f(A)) = f^{-1}(B) + A$.

The following is a useful application of this,

Proposition 3.2: If A is a submodule of M and B is a submodule of N then

- a. $f(f^{-1}(B)) = B \cap \text{Im } f$;
- b. $f^{-1}(f(A)) = A + \text{Ker } f$.

It is useful to remember certain facts about composition.

Lemma 3.4:

- a. if $f : M \rightarrow N$ and $g : N \rightarrow P$ are R -morphisms then so is $g \circ f$;
- b. if $f : M \rightarrow N$ and $g : N \rightarrow P$ are R -monomorphisms then so is $g \circ f$;
- c. if $f : M \rightarrow N$ and $g : N \rightarrow P$ are R -epimorphisms then so is $g \circ f$;
- d. if $g \circ f$ is an epimorphism then so is g ;
- e. if $g \circ f$ is a monomorphism then so is f .

With respect to composition, we are interested in “diagram completing” problems. That is, suppose we have a diagram of the form

$$\begin{array}{ccc} A & \xrightarrow{g} & C \\ \downarrow f & & \\ B & & \end{array}$$

and we want to know the conditions under which there exists an R -morphism $h : B \rightarrow C$ such that $h \circ f = g$. We also have the “dual” problem, where we have a diagram of the form

$$\begin{array}{ccc} & C & \\ & \downarrow g & \\ B & \xrightarrow{f} & A \end{array}$$

and we want to know the conditions under which there exists an R -morphism $h : C \rightarrow B$ such that $f \circ h = g$. As a warmup, we consider these problems when A, B, C are sets and f, g are simply mappings.

Proposition 3.3:

- a. If A, B, C are non-empty sets and $f : A \rightarrow B$ and $g : A \rightarrow C$ are mappings then the following statements are equivalent
 - there exists a mapping $h : B \rightarrow C$ such that $h \circ f = g$;
 - $\forall x, y \in A. f(x) = f(y) \Rightarrow g(x) = g(y)$.
- b. If A, B, C are non-empty sets and $f : B \rightarrow A$ and $g : C \rightarrow A$ are mappings then the following statements are equivalent
 - there exists a mapping $h : C \rightarrow B$ such that $f \circ h = g$;
 - $\text{Im } g \subseteq \text{Im } f$.

4. Quotient modules and isomorphism theorems

5. Chain conditions and Jordan-Hölder towers